Software validation report

Record of the validation of the data-processing software before use and after changes (ISO/IEC 17025:2017 7.11.2). Every reference example is recomputed from its inputs and compared with the published values within a documented tolerance.

Software	KAPSAM
Version	0.1.1 (commit 951eb6886563, 2026-11-11)
Compiler	rustc 1.97.0 (2d8144b78 2026-07-07)
Platform	linux x86_64
Date	2026-11-11
Options	Monte Carlo evaluations included (fixed seeds and numbers of trials as stored in the files)
Digest of the checked files	b598cb226808526e510c0d605dd552da034d6b88931071bae3d4a7dc7c50829d (SHA-256 of the sorted list "SHA-256 file" of all checked files; each file holds its inputs, reference values and settings (seeds, numbers of trials, rounding))

Summary

284 project files and 1843 compared values. Passed: 284; failed: 0; without reference values: 0.

All reference values are reproduced within their tolerances.

Results by method

Method	Projects	Values	Outcome
top-down	39	165	passed
GUM/MCM	132	505	passed
JCGM 106, multicomponent	1	7	passed
ISO 6143	8	87	passed
JCGM 106	20	70	passed
ISO 13528	9	183	passed
JCGM 102	17	206	passed
ISO 5725-2	6	163	passed
consensus (NIST Consensus Builder)	9	175	passed
Nordtest TR 569	22	98	passed
Eurachem UfS:2019	7	61	passed
EURAMET cg-18 App. C	9	103	passed
GUM/MCM + JCGM 106	5	20	passed

Results by project: reference examples

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
bettencourt2026-fluoranthene-aqueous	R. J. N. Bettencourt da Silva, Section 3.2; Supplementary Material 06, sheets Trueness, Performance Model and Analysis sequence	13	0.049	passed	97e940f8bcb0
bettencourt2026-fluoranthene-weighted	R. J. N. Bettencourt da Silva, Section 3.2; Supplementary Material 06, sheets Trueness, Performance Model and Analysis sequence	13	0.048	passed	d7d76dfc0e60
bipm-isrd-01-qnmr-dmso2-ma-d20	BIPM, Table 4	2	0.68	passed	12f2b939fe63
bipm-isrd-01-qnmr-dmtp-ma-dmso-d6	BIPM, Table 6	2	0.86	passed	0bc8eda567b9
bipm-isrd-02-qnmr-dmtp-dmso2-d6-m3	BIPM, Table 6	2	0.23	passed	d850f5e5e799

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
bipm-isrd-02-qnmr-ma-dmso2-d6	BIPM, Table 4	2	0.38	passed	c8ab925f8132
bipm-isrd-06-qnmr-ba-btmsb-d4-acn	BIPM, Table 6	9	0.33	passed	5e8f1ae4bf11
bipm-isrd-06-qnmr-dmso2-btmsb-d4-acn	BIPM, Table 6	2	0.24	passed	3d5516ee2423
bipm-isrd-07-qnmr-dmso2-dss-d6-d2o	BIPM, Table 4	2	0.75	passed	6ae038663a3f
bipm-isrd-07-qnmr-ma-dss-d6-d2o	BIPM, Table 6	2	0.44	passed	087ed8e31831
bipm-isrd-08-qnmr-btmsb-d4-bad-d2o	BIPM, Table 2	2	0.31	passed	e9b31c2d4d82
emue-e2.1-multicomponent-mediation	EMPIR 17NRM05 EMUE, Example E2.1, Tables E2.1.1–E2.1.2, E2.1.6	7	0.45	passed	eb481dbe7beb
emue-e3.3-no2-analyser-dynamic-dilution	EMPIR 17NRM05 EMUE, Example E3.3, Tables E3.3.2–E3.3.5, (E3.3.17)–(E3.3.19)	4	0.38	passed	bd893c0d6eaa
emue-e3.3-no2-analyser-dynamic-dilution-v2	EMPIR 17NRM05 EMUE, Example E3.3, Tables E3.3.2–E3.3.5, (E3.3.17)–(E3.3.19)	7	0.38	passed	bcd85ed3a332
emue-e3.4-soil-lead-iso21748-mass-bias	EMPIR 17NRM05 EMUE, E3.4.6–E3.4.7, Tables E3.4.1–E3.4.7, equation (E3.4.18)	3	0.30	passed	d1725afa02ae
emue-e3.4-soil-lead-iso21748	EMPIR 17NRM05 EMUE, E3.4.6–E3.4.7, Tables E3.4.1–E3.4.7, equation (E3.4.18)	3	0.46	passed	0442de5e853f
emue-e3.8-ammonia-permeation	EMPIR 17NRM05 EMUE, Example E3.8, (E3.8.1)–(E3.8.4), Tables E3.8.3–E3.8.5	5	0.11	passed	a75bd9bb7046
emue-e6.5-ephedrine-wada-decision-limit	EMPIR 17NRM05 EMUE, Example E6.5, E6.5.2–E6.5.3, Table E6.5.2	4	0.27	passed	a6212175abc9
emue-e6.5-ephedrine-wada-topdown	EMPIR 17NRM05 EMUE, Example E6.5, Table E6.5.1	3	0.42	passed	47870398395e
eurachem-compliance-b1-nickel-steel	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 1; Annex A, Case 1b	4	0.45	passed	bbea091120fb
eurachem-compliance-b2-upper-limit	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 2; Annex A, Case 2	3	0.20	passed	e98c6dc46045
eurachem-compliance-b3-banned-substance-exact	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 3; Annex A, Case 4 and Table 1	3	0.43	passed	8c3fcae894df
eurachem-compliance-b3-banned-substance-mid	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 3; Annex A, Case 4 and Table 1	3	0.46	passed	a691bc80be45
eurachem-compliance-b3-banned-substance-tail	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 3; Annex A, Case 4 and Table 1	3	0.45	passed	122a58cace7f
gum5-e1-ph-temperature-interpolation-5.2026	JCGM GUM-5:2026, Example 1, 1.6, 1.9 and 1.10.3	4	0.95	passed	f00bd91918ec
gum5-e1-ph-temperature-interpolation	JCGM GUM-5:2026, Example 1, 1.6, 1.9 and 1.10.3	4	0.95	passed	bd687dafbcc6
gum5-e1-ph-two-point-25c-corrected-standards	JCGM GUM-5:2026, Example 1, 1.10.3.4 and 1.10.3.5	2	0.65	passed	399dfe74085e
gum5-e1-ph-two-point-25c	JCGM GUM-5:2026, Example 1, 1.8	4	0.55	passed	a46f9dc42415
gum5-e10-tspm-global-risks-acceptance-0.17	JCGM GUM-5:2026, Example 10, 10.6.3	3	0.17	passed	196e2ebbf055
gum5-e10-tspm-global-risks-acceptance-0.22	JCGM GUM-5:2026, Example 10, 10.6.3	3	0.19	passed	a5e6abf8a3d3
gum5-e10-tspm-global-risks-shared-risk	JCGM GUM-5:2026, Example 10, 10.6.3	3	0.14	passed	ba9d50ed7edc
gum5-e10-tspm-specific-risks	JCGM GUM-5:2026, Example 10, 10.6.4 and 10.7.2	4	0.18	passed	557f0ccf9176
gum5-e12-between-bottle-homogeneity	JCGM GUM-5:2026, Example 12, Tables 12.1–12.3, 12.4.5 and 12.4.8	14	0.88	passed	9678e5ee9175
gum5-e2-benzo-a-pyrene-ensemble	JCGM GUM-5:2026, Example 2, Tables 2.1–2.3; Willink, Metrologia 44 (2007) 340, §4.1.1	3	0.57	passed	12a36c83cdc4

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
gum5-e2-benzo-a-pyrene	JCGM GUM-5:2026, Example 2, Tables 2.1–2.3	4	0.82	passed	4ceffd1cfb36
gum5-e3-molar-mass-glucose	JCGM GUM-5:2026, Example 3, Tables 3.1 and 3.2	2	0.20	passed	d7e7f39a7c87
gum5-e4-gravimetric-gas-mixture	JCGM GUM-5:2026, Example 4, Tables 4.1–4.6 and 4.3.6.9	19	0.79	passed	5461bf527cc5
gum5-e5-ghg-inventory-agriculture-manured-arable	JCGM GUM-5:2026, Example 5, Tables 5.1–5.3 and 5.4.9	4	0.95	passed	9520645f3e4c
gum5-e5-ghg-inventory-agriculture	JCGM GUM-5:2026, Example 5, Tables 5.1–5.3 and 5.4.9	4	0.95	passed	afa1b4539bfb
iso13528-e1-censored-deleted	ISO 13528:2022, Annex E.1 (Table E.1)	2	0.50	passed	d79653d9ec97
iso13528-e1-censored-half	ISO 13528:2022, Annex E.1 (Table E.1)	2	0.85	passed	4a0e98d3bac8
iso13528-e1-censored-ignored	ISO 13528:2022, Annex E.1 (Table E.1)	2	0.18	passed	3c6267ae0c18
iso13528-e2-arsenic-homogeneity	ISO 13528:2022, E.2, Tables E.2 and E.3	8	0.65	passed	10f1f474c31c
iso13528-e3-atrazine-algorithm-a	ISO 13528:2022, E.3, Tables E.4 and E.5	13	0.84	passed	a5a9f01d4db0
iso13528-e3-atrazine-median	ISO 13528:2022, E.3, Table E.5	6	0.84	passed	646d4f79bdc4
iso13528-e4-mercury-feed	ISO 13528:2022, E.4, Tables E.6 and E.7	131	0.98	passed	f80274289f87
iso13528-e5-los-angeles-reference-value	ISO 13528:2022, Annex E.5 (Table E.8)	2	0.69	passed	fb8e1c1f2fa8
iso21748-c2-meat-content	ISO 21748:2017, Annex C.2, Table C.1	3	0.74	passed	e16b77cafad5
iso21748-c4-crude-fibre	ISO 21748:2017, Annex C.4, Tables C.6–C.9	6	0.99	passed	f7a71fc80945
iso5725-2-c1-sulfur-in-coal-reml	ISO 5725-2:2019, Annex C.1.9 (Table C.6), Annex B.2	16	0.91	passed	3401dfa58482
iso5725-2-c1-sulfur-in-coal	ISO 5725-2:2019, Annex C.1 (Tables C.1–C.5, Formulae C.1–C.24)	36	0.99	passed	62104ec9cf9d
iso5725-2-c3-creosote-oil-reml	ISO 5725-2:2019, Annex C.3.8 (Table C.19), Annex B.2	20	0.95	passed	20909b80d171
iso5725-2-c3-creosote-oil	ISO 5725-2:2019, Annex C.3 (Tables C.14–C.18) and 8.5 (Tables 1–4)	44	1.0	passed	fa32b0832d3c
iso5725-3-j1-carbon-steel	ISO 5725-3:2023, Annex J.1, Table J.1	1	0.13	passed	48e94c8cebe0
iso5725-3-j2-vanadium-steel	ISO 5725-3:2023, Annex J.2, Tables J.2–J.5	1	0.12	passed	6872326e58b8
iso6143-b1-no-uv-linear	ISO 6143:2025, Annex B.2.2 (Tables B.1–B.6)	14	0.62	passed	144afd0e97e2
iso6143-b2-n2-natural-gas-covariances	ISO 6143:2025, Annex B.2.3 (Table B.10)	9	0.29	passed	5eeb4fbcfe0a
iso6143-b2-n2-natural-gas-linear	ISO 6143:2025, Annex B.2.3 (Tables B.7, B.8)	12	0.29	passed	15952e9c1ddd
iso6143-b2-n2-natural-gas-quadratic	ISO 6143:2025, Annex B.2.3 (Table B.9)	16	0.24	passed	35e7e507d216
iso6143-b3-no-ndir-exponential	ISO 6143:2025, Annex B.1 and B.2.4 (Tables B.11, B.12)	11	0.028	passed	de9178cb3724
iso6143-b3-no-ndir-power	ISO 6143:2025, Annex B.2.4 (Tables B.11, B.12)	14	0.64	passed	2f4905126e01
iupac2024-3a-argentometric-titration-lo	Possolo et al., Examples 3A and 4I; Meija et al., Eq. (19)	2	0.10	passed	23a70b037582
iupac2024-3e-4m-iridium-molar-mass-ave	Possolo et al., Examples 3E, 4M, 4N	4	0.80	passed	83d35ae39d79
iupac2024-4a-water-hardness	Possolo et al., Examples 3C and 4A	2	0.89	passed	f26e42a6bf3a
iupac2024-4b-silicon-isotopic-abundances	Possolo et al., Example 4B	8	0.90	passed	020f0b992f1e
iupac2024-4f-ccqm-k145-nickel-bovine-liver	Possolo et al., Example 4F	4	0.92	passed	191f2fd219cf
iupac2024-4h-cadmium-standardization	Possolo et al., Example 4H	6	0.73	passed	676f526bdc77
iupac2024-4i-argentometric-titration	Possolo et al., Examples 3A and 4I	4	0.77	passed	1ac5d70808eb
iupac2024-5c-benzoic-acid-purity-beta	Possolo et al., Example 5C	4	0.48	passed	7c79d40af518
mupr-b1-nitrate	Eurachem/CITAC Guide, Example B1	7	0.96	passed	14bd744fe2ea

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
mupr-b2-pooled-precision	Eurachem/CITAC Guide, Example B2, Tables B2.1–B2.3	6	0.50	passed	acf4f4c5f51a
mupr-b2-replicates-on-two-days	Eurachem/CITAC Guide, Example B2, Eq. (B2.8)–(B2.9)	1	0.051	passed	92b534629cc0
mupr-b3-linear-precision	Eurachem/CITAC Guide, Example B3, Table B3.1, Eq. (B3.1)	1	0.82	passed	1aac173149a7
mupr-b4-arsenic-one-crm	Eurachem/CITAC Guide, Example B4	9	0.98	passed	157dea4a0d3a
mupr-b5-crm-and-spiked-samples	Eurachem/CITAC Guide, Example B5	7	0.83	passed	0621d638a70c
mupr-b6-sediment-22-pt	Eurachem/CITAC Guide, Example B6, Tables B6.1–B6.3	11	0.93	passed	36b33cb67a77
mupr-ex2-reference-materials	Eurachem/CITAC Guide, Example 2	3	0.36	passed	d22128167278
mupr-ex3-spiked-samples	Eurachem/CITAC Guide, Example 3	2	0.39	passed	9d0fc8aa8b30
nicob-pcb28-sediment-degrees-of-freedom	NIST Consensus Builder user manual (Koepke, §4.2, Exhibits 6 and 7; §A.2, §A.4, §A.5.1	36	0.65	passed	4496e92490a8
nist-tn1900-e8-molecular-weight-s2	NIST Technical Note 1900, Example E8, Exhibit 12	5	0.36	passed	c4d30cb80a2b
nist-tn1900-e9-cadmium-standard	NIST Technical Note 1900, Example E9	5	0.73	passed	461267d7bd72
nordtest-3-ammonium-pt	Nordtest TR 537 ed. 4 (2017), §3.2 and Appendix 4	7	0.94	passed	2cfd4e952b23
nordtest-5.3-ammonium-duplicates	Nordtest TR 537 ed. 4 (2017), §5.3 and Appendix 5	4	0.65	passed	44572cd9b0f3
nordtest-5.3-oxygen-duplicates	Nordtest TR 537 ed. 4 (2017), §5.3 and Appendix 6	2	0.93	passed	818066c0b900
nordtest-6.1-one-crm	Nordtest TR 537 ed. 4 (2017), §6.1	3	0.35	passed	fd25680700ef
nordtest-6.1-three-crms	Nordtest TR 537 ed. 4 (2017), §6.1	3	0.51	passed	7a90650eb056
nordtest-6.2-proficiency-tests	Nordtest TR 537 ed. 4 (2017), §6.2	3	0.71	passed	b0369555b563
nordtest-6.3-recovery	Nordtest TR 537 ed. 4 (2017), §6.3	3	0.57	passed	0ab762c2219d
nordtest-8-bod-crm	Nordtest TR 537 ed. 4 (2017), §8.2 Example A, Appendix 7	4	0.80	passed	5efed90d1d68
nordtest-8-bod-pt	Nordtest TR 537 ed. 4 (2017), §8.2 Example B, Table 6	5	0.97	passed	b9e9c4f2cff6
nordtest-8-pcb-crm	Nordtest TR 537 ed. 4 (2017), §8.3 Example C	3	0.76	passed	05ee0aca2326
nordtest-8-pcb-pt	Nordtest TR 537 ed. 4 (2017), §8.3 Example D	5	0.61	passed	52dacc1395be
palma2019-cr-marine-sediment	C. Palma, Table 9	2	0.11	passed	3bbc5811d4ea
palma2019-cu-marine-sediment	C. Palma, Table 9	2	0.75	passed	35f8f04d9226
palma2019-li-marine-sediment	C. Palma, Table 9	2	0.30	passed	8fa0b5f6e48a
palma2019-mn-marine-sediment	C. Palma, Table 9	2	0.22	passed	49607ffc6cd9
palma2019-ni-marine-sediment	C. Palma, Table 9	2	0.33	passed	11311487a5e7
palma2019-pb-marine-sediment	C. Palma, Table 9, Section 4.1	2	0.49	passed	f287368eec64
palma2019-zn-marine-sediment	C. Palma, Table 9	2	0.46	passed	48110fb13ded
possolo-bodnar-2018-chromium-isotopic-ratio	Possolo & Bodnar, §4.4, Tables 3 and 4	5	0.89	passed	0c4aefff8bfc
possolo-meija-2022-atomic-weight-lead-pb1	Possolo & Meija, Sums, Products, and Ratios: Atomic Weight of Lead, pp. 26–29	12	0.97	passed	19419876615a
possolo-meija-2022-plasma-anion-gap	Possolo & Meija, Sums, Products, and Ratios: Plasma Anion Gap, pp. 16–18	2	0.35	passed	491b7c71b971
quam-a1-cadmium-standard-volume	EURACHEM/CITAC Guide CG 4, Example A1, A1.4	2	0.59	passed	6e5e9c698534
quam-a1-cadmium-standard	EURACHEM/CITAC Guide CG 4, Example A1, Tables A1.2 and A1.3	3	0.38	passed	8a1981901836

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
quam-a2-naoh-standardisation-sucrose	EURACHEM/CITAC Guide CG 4, Example A2, A2.4	2	0.32	passed	88cc0255b152
quam-a2-naoh-standardisation	EURACHEM/CITAC Guide CG 4, Example A2, Tables A2.2 and A2.3	3	0.80	passed	267a7d47140a
quam-a3-hcl-titration	EURACHEM/CITAC Guide CG 4, Example A3, Tables A3.2 and A3.3	4	0.69	passed	0b7c591d630a
quam-a4-pesticides-combination	EURACHEM/CITAC Guide CG 4, Example A4, Tables A4.4 and A4.5	3	0.48	passed	c9de239e7d98
quam-a4-pesticides-topdown	EURACHEM/CITAC Guide CG 4, Example A4, Tables A4.2–A4.5	7	0.91	passed	1aabd2867018
quam-a5-cadmium-release	EURACHEM/CITAC Guide CG 4, Example A5, Tables A5.3 and A5.4	4	0.86	passed	ba9c48b7d745
quam-a6-crude-fibre	EURACHEM/CITAC Guide CG 4, Example A6, Tables A6.4–A6.5	6	0.43	passed	0ec5386edcd1
quam-a7-assay-standard	EURACHEM/CITAC Guide CG 4, Example A7, Eq. (8), Table A7.3	2	0.46	passed	a767a11a8280
quam-a7-lead-double-idms	EURACHEM/CITAC Guide CG 4, Example A7, Eq. (5), Tables A7.4–A7.8	2	0.16	passed	23920f968c3d
tr569-ch7-case1-statistical	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limits and the central line in X-charts, case 1	4	0.72	passed	a010c1426afd
tr569-ch7-case2-target-mean	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limits and the central line in X-charts, case 2	4	0.40	passed	7141d66120c1
tr569-ch7-case3-target-reference	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limits and the central line in X-charts, case 3	4	0	passed	474850bed722
tr569-ch7-range-case1-statistical	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limit in R or r%-charts, case 1	2	0.24	passed	77ba5783baaf
tr569-ch7-range-case2-target	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limit in R or r%-charts, case 2	3	0.33	passed	20c2abbf6675
tr569-ex1-nickel-steel-xrf	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 1	5	0.52	passed	70b1c501730a
tr569-ex10-boron-toys-target	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 10	4	0	passed	5f87ef68ba91
tr569-ex12-pooling-sr-srw	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 12 and footnote 24	5	0.94	passed	a404e9ec7134
tr569-ex2-cobalt-steel-xrf	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 2	4	2.8×10^{-12}	passed	63859fea3d4d
tr569-ex3-ammonium-r	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 3	4	0.80	passed	e52d80ca7b97
tr569-ex3-ammonium-x	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 3	4	0.60	passed	00e3c1ff64dd
tr569-ex4-lead-water-icpms	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 4	4	1.1×10^{-13}	passed	dc2e963ef0d6
tr569-ex5-arsenic-crm-tolerance	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 5	2	0	passed	67a32af10def
tr569-ex6-phosphorus-relative-range	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 6	3	0.91	passed	ebfac2b11550
tr569-ex7-beta-hch-crm-target	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 7	5	0	passed	683e4066fa05
tr569-ex8-copper-r-review	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 8	6	0.76	passed	933d19ab852e

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
tr569-ex8-copper-x-new-limits	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 8	4	0.92	passed	9782b6d52166
tr569-ex8-copper-x-review	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 8	11	0.98	passed	ecec550b6284
tr569-ex9-zinc-procedural-blank	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 9	4	2.8×10^{-8}	passed	32389aea4393
tr569-table1-zinc	Nordtest TR 569 ed. 6 (2026), Chapter 2, Table 1 and Figure 1; Chapters 7 and 9	9	0.99	passed	9d4d05db4a66
ufs-a1-nitrate-lettuce	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A1, Table A1.1 and Figure A1.2	8	0.60	passed	b973c0f9827e
ufs-a2-lead-soil-uncertainty-factor	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A2, Tables A2.2–A2.4 and Figure A2.2	22	0.73	passed	d489ca48b672
ufs-a3-iron-groundwater-anova	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A3, Tables A3.5 and A3.7	3	0.42	passed	c66a94d4ddb1
ufs-a3-iron-groundwater-range	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A3, Tables A3.4 and A3.7	8	0.87	passed	469a337bd830
ufs-a4-vitamin-a-porridge-40g	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A4, Tables A4.3, A4.5–A4.7, A4.14, A4.15 and summary	7	0.76	passed	0675d43f51af
ufs-a4-vitamin-a-porridge-4g	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A4, Tables A4.4, A4.8–A4.10	5	0.80	passed	ae944956c424
ufs-a4-vitamin-a-qc-chart	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A4, Section 7.3, Table A4.13 and Figure A4.4; Section 13.2	5	0.86	passed	594e4283789d
ufs-a5-enzyme-feed-gy	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A5, Table A5.1; Section 10.2.7–10.2.8, Equations 6–8	8	0.96	passed	aef4031afda2
ufs-a6-cadmium-soil	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A6, Tables A6.1–A6.6	2	0.23	passed	58326ca9bce1
ufs-a6-phosphorus-soil	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A6, Tables A6.1–A6.6	2	0.40	passed	9374b5c1c6b5
ea402-s3-resistor-10kohm	EA-4/02 M:2022, S3	3	0.69	passed	1745db64f568
ea402-s6-power-sensor-18ghz	EA-4/02 M:2022, S6	4	0.71	passed	e6f4998f2942
ea402-s7-step-attenuator-30db	EA-4/02 M:2022, S7	3	0.94	passed	f467f5736cad
ea402-s9-dmm-100v	EA-4/02 M:2022, S9	2	0.85	passed	8974827a9eeb
gum-h2-impedance	JCGM 100:2008, H.2, Tables H.2 and H.3	2	0.67	passed	18c8a39d92ed
gum-h2-reactance	JCGM 100:2008, H.2, Tables H.2 and H.3	2	0.98	passed	2f502deaa633
gum-h2-resistance	JCGM 100:2008, H.2, Tables H.2 and H.3	2	0.34	passed	ecb068cde7db
jcgm101-9.4-comparison-loss-x000	JCGM 101:2008, 9.4, Table 9	6	0.093	passed	0d39533acfa2
jcgm101-9.4-comparison-loss-x001	JCGM 101:2008, 9.4, Table 9	5	0.093	passed	211049f31296
jcgm101-9.4-comparison-loss-x002	JCGM 101:2008, 9.4, Table 8	6	0.042	passed	bd63d1e33a71
jcgm101-9.4-comparison-loss-x003	JCGM 101:2008, 9.4, Table 9	6	0.065	passed	ff624422b378
jcgm101-9.4-comparison-loss-x004	JCGM 101:2008, 9.4, Table 9	5	0.065	passed	14931cc6adee
jcgm101-9.4-comparison-loss-x005	JCGM 101:2008, 9.4, Table 8	6	0.39	passed	7024d3f649e9

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
jcgm101-9.4-comparison-loss-x001-correlated	JCGM 101:2008, 9.4, Table 9	6	0.22	passed	620ce851584f
jcgm101-9.4-comparison-loss-x010-correlated	JCGM 101:2008, 9.4, Table 9	5	0.22	passed	495dc8086d43
jcgm101-9.4-comparison-loss-x100	JCGM 101:2008, 9.4, Table 8	6	0.99	passed	73e10b92a42f
jcgm102-9.3-polar-x001-correlated	JCGM 102:2011, 9.3 (Table 7)	10	0.43	passed	a25926dbcb0f
jcgm102-9.3-polar-x001	JCGM 102:2011, 9.3 (Table 6)	10	0.71	passed	fec5d6261ba6
jcgm102-9.3-polar-x010-correlated	JCGM 102:2011, 9.3 (Table 7)	10	0.49	passed	440ac9da1b5f
jcgm102-9.3-polar-x010	JCGM 102:2011, 9.3 (Table 6)	10	0.82	passed	e78f4f5447d3
jcgm102-9.3-polar-x100-correlated	JCGM 102:2011, 9.3 (Table 7)	10	0.81	passed	2faa0ee7654b
jcgm102-9.3-polar-x100	JCGM 102:2011, 9.3 (Table 6)	10	0.83	passed	693701192151
jcgm102-9.4-resistance-reactance-correlated	JCGM 102:2011, 9.4 (Tables 8–11)	12	0.98	passed	0fe1e6a3bc67
jcgm102-9.4-resistance-reactance	JCGM 102:2011, 9.4 (Tables 8–11)	12	0.98	passed	28d9e5b318e2
m3003-c5-dvm-dominant-rectangular	UKAS M3003 ed. 6 (2024), Appendix C.5–C.10	4	0.55	passed	e56eb32e70b1
m3003-k1-resistor-10kohm	UKAS M3003 ed. 6 (2024), Appendix K.1	3	0.70	passed	36195ab2105a
m3003-k2-power-sensor-18ghz	UKAS M3003 ed. 6 (2024), Appendix K.2	3	1.0	passed	b17ec2b7bcc0
m3003-k3-attenuator-30db-mismatch	UKAS M3003 ed. 6 (2024), Appendix K.3 and E.9.5	3	1.7×10^{-11}	passed	692b9150fbde
m3003-k3-attenuator-30db	UKAS M3003 ed. 6 (2024), Appendix K.3	3	0.99	passed	b466a13f58f9
m3003-l3-dmm-1v-range-095v	UKAS M3003 ed. 6 (2024), Appendix L.3	3	0.53	passed	60e74c14aee6
m3003-l3-dmm-1v-range-1v-dominant	UKAS M3003 ed. 6 (2024), Appendix L.3	2	0.12	passed	097a8c15a007
m3003-l3-dmm-1v-range-500mv	UKAS M3003 ed. 6 (2024), Appendix L.3	2	0.51	passed	8b4de14783fb
nist-tn1900-e11-step-attenuator	NIST Technical Note 1900, Example E11, Exhibits 16 and 17	4	0.64	passed	879177cf3760
num-10-resistance	JCGM 100:2008, H.2, Tables H.2 and H.3	6	0.43	passed	b5f9b3be7398
possolo-meija-2022-wheatstone-bridge	Possolo & Meija, Sums, Products, and Ratios: Wheatstone Bridge, pp. 24–26	2	0.084	passed	1635e714f4a5
cg19-ex1-flask-1000ml	EURAMET Calibration Guide No. 19 v4.1 (2025), 7.1 (Tables 1 and 2)	5	0.63	passed	1fbd3c1e8d4d
cg19-ex2-pycnometer-100ml	EURAMET Calibration Guide No. 19 v4.1 (2025), 7.2 (Tables 3 and 4)	5	0.85	passed	a0e2abe4e822
ea402-s12-water-meter-error	EA-4/02 M:2022, S12	2	0.59	passed	26d6ce337d68
ea402-s12-water-meter-mean-error	EA-4/02 M:2022, S12	3	0.74	passed	a2fdf31ca332
ea402-s12-water-meter-volume	EA-4/02 M:2022, S12	2	0.60	passed	e9b27f4329ab
m3003-k9-flowrate-standard-condition	UKAS M3003 ed. 6 (2024), Appendix K.9	4	0.90	passed	eb4e1f812383
nist-tn1900-e4-pitot-tube	NIST Technical Note 1900, Example E4	5	0.68	passed	bb91f86703b1
gum5-e9-torque-slope-ols	JCGM GUM-5:2026, Example 9, Tables 9.1–9.3	4	0.91	passed	d4e0c049019b
gum5-e9-torque-slope-wls	JCGM GUM-5:2026, Example 9, Tables 9.1–9.3	4	0.82	passed	e57e91bff780
iso21748-c1-co-emissions	ISO 21748:2017, Annex C.1	3	0	passed	ae10d6471462
iso21748-c3-aerobic-plate-count-2l	ISO 21748:2017, Annex C.3, Tables C.2–C.4	3	0.54	passed	43a5bd2dbfa9
iso21748-c3-aerobic-plate-count-2mp	ISO 21748:2017, Annex C.3, Tables C.2–C.4	3	0.85	passed	48f3292cd03a
iso21748-c3-aerobic-plate-count-vegetable	ISO 21748:2017, Annex C.3, Tables C.2–C.4	3	0.73	passed	dce6e6671de2
jcgm101-9.2.2-additive-gaussian	JCGM 101:2008, 9.2.2	7	0.25	passed	61c10608eb3f
jcgm101-9.2.3-additive-rectangular	JCGM 101:2008, 9.2.3	6	0.29	passed	481a95e5b4f7
jcgm101-9.2.4-additive-dominant-rectangular	JCGM 101:2008, 9.2.4	5	0.33	passed	e6df7f442a4b
jcgm102-9.2.2-additive-gaussian	JCGM 102:2011, 9.2.2 (Table 3)	16	0.45	passed	6af1973c3a6d
jcgm102-9.2.3-additive-rectangular	JCGM 102:2011, 9.2.3 (Table 4)	16	0.45	passed	d9fde9850945
jcgm102-9.2.4-additive-dominant-rectangular	JCGM 102:2011, 9.2.4 (Table 5)	16	0.56	passed	5a31c054deb4

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
jcgm106-7.3-bursting-strength	JCGM 106:2012, 7.3.3, Example 2	2	0.20	passed	8edb9ef79fbe
jcgm106-7.3-zener-diode	JCGM 106:2012, 7.3.3, Example 1	2	0.15	passed	5a8dceaa8573
jcgm106-7.4-motor-oil-viscosity	JCGM 106:2012, 7.4, Example	3	0.53	passed	eaff9c8aff4f
jcgm106-8.3-nandrolone	JCGM 106:2012, 8.3.3, Example 2	1	0.68	passed	1eb4125b4357
jcgm106-8.3-speed-limit	JCGM 106:2012, 8.3.3, Example 1	3	0.82	passed	3659745800c1
jcgm106-9.5-ball-bearings	JCGM 106:2012, 9.5.4, Example	5	0.76	passed	d7dcbb06ca16
jcgm106-9.5-precision-resistors	JCGM 106:2012, 9.5.3, Example	3	0.88	passed	7adcf527bba4
num-3-ratio-abc	T. Lafarge, §3, Figure 1	7	0.92	passed	9b76186b1311
tr569-ex11-micropipette-tolerance	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 11	2	0	passed	653424ca3244
tur-guard-band-dobbert-tur2-5	JCGM 106:2012, 9.5 (global risks)	6	3.7×10^{-9}	passed	b8c7ccd2af0e
tur-guard-band-rp10-tur2-0	JCGM 106:2012, 9.5 (global risks)	6	3.9×10^{-9}	passed	7ded1f140c67
tur-guard-band-rss-tur1-5	JCGM 106:2012, 9.5 (global risks)	6	2.7×10^{-9}	passed	66675e8b92c4
ea402-s10-vernier-calliper	EA-4/02 M:2022, S10	2	0.68	passed	ffc45534bcb4
ea402-s13-ring-gauge-90mm	EA-4/02 M:2022, S13	3	0.70	passed	4a6774391883
ea402-s13-ring-gauge-temperature	EA-4/02 M:2022, S13	2	0.40	passed	397ae464920a
ea402-s4-gauge-block-50mm	EA-4/02 M:2022, S4	3	0.92	passed	b3ea19f34d3d
gum-h1-end-gauge	JCGM 100:2008, H.1, Table H.1	5	0.83	passed	de2a2c7e90e5
jcgm101-9.5-gauge-block-approximate	JCGM 101:2008, 9.5, Tables 10 and 11	7	0.35	passed	0793c596b97d
jcgm101-9.5-gauge-block-nonlinear	JCGM 101:2008, 9.5, Tables 10 and 11	7	0.35	passed	1ede84b3bbf2
m3003-k10-sheet-thickness	UKAS M3003 ed. 6 (2024), Appendix K.10	4	0.92	passed	7788b0ac7948
m3003-k6-gauge-block-10mm-temperature	UKAS M3003 ed. 6 (2024), Appendix K.6	3	0.068	passed	e16dd56d1329
m3003-k6-gauge-block-10mm	UKAS M3003 ed. 6 (2024), Appendix K.6	3	0.71	passed	85d98bbc4910
num-8-end-gauge-gaussian	T. Lafarge, §8, Table 3 and Figure 3	7	0.50	passed	a8b3a86a0fda
possolo-meija-2022-storage-tank-volume	Possolo & Meija, Sums, Products, and Ratios: Storage Tank, pp. 18–24	4	0.63	passed	aaeeb0f070b8
cg18-h1-100g-option2	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	4	0.97	passed	41f3b133f94f
cg18-h1-220g-max-option1-delta-L	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	5	0.65	passed	322a870fbac5
cg18-h1-220g-max-option1	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	4	0.39	passed	7d380540f128
cg18-h1-220g-max-option2	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	4	0.82	passed	5154c8d16e91
cg18-h1-220g-zero	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	5	0.77	passed	6dd4711c2234
cg18-h1-a-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1.1/A to H1.4/A; 7.4, 7.5; Appendix G	8	0.28	passed	fe1df4df2d7f
cg18-h1-b-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1.1/B to H1.4/B; 7.4, 7.5; Appendix G	8	0.71	passed	d5dada5893c4
cg18-h2-60kg-10kg-option1	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	5	0.96	passed	d897ae464801
cg18-h2-60kg-max-adjusted	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/B	5	0.96	passed	e5419b841ea0

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
cg18-h2-60kg-max-option1-delta-U	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	4	0.90	passed	621546a5a299
cg18-h2-60kg-max-option1	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	5	0.93	passed	990abcc77a74
cg18-h2-60kg-max-option2	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	4	0.89	passed	b74c636cad3d
cg18-h2-60kg-zero	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	4	0.77	passed	d838031a9642
cg18-h2-a-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2.1/A to H2.4/A; 7.4, 7.5; Appendix G	19	0.30	passed	d5cb624eac61
cg18-h2-b-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2.1/B to H2.4/B; 7.4, 7.5; Appendix G	19	0.47	passed	c3f7350afe3c
cg18-h3-30t-10t-weights	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3, H3.2/A and H3.3/A	5	0.97	passed	2e8ca2acb3f3
cg18-h3-30t-15t-substitution	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3, H3.2/A and H3.3/A	2	0.19	passed	f2d8228406ec
cg18-h3-30t-max-substitution	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3, H3.2/A and H3.3/A	4	0.81	passed	e07e337fcebe
cg18-h3-30t-zero	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3, H3.2/A and H3.3/A	4	0.27	passed	cd9c318ba12d
cg18-h3-a-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3.1/A to H3.4/A; 7.4, 7.5; Appendix G	9	0.87	passed	e45f0947730d
cg18-h3-b-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3.1/B to H3.4/B; 7.4, 7.5; Appendix G	9	0.67	passed	f4ae0f60adf2
cg18-h4-400g-error-function-smeared	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H4; Appendix C2.2	14	0.97	passed	b5855bca1502
cg18-h4-400g-error-function-smear25	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H4; Appendix C2.2	13	0.90	passed	2dda74a14c9e
cg18-h4-400g-error-function	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H4; Appendix C2.2	4	0.81	passed	87200041a343
ea402-s2-weight-10kg	EA-4/02 M:2022, S2	3	0.87	passed	46f6088c84be
ilac-p14-cmc-floor-weight-10kg	EA-4/02 M:2022, S2	4	0.87	passed	69c5c7a99c61
jcgm101-9.3-mass-calibration-hybrid-model	JCGM 101:2008, 9.3, Table 6	6	0.73	passed	675cb13666e9
jcgm101-9.3-mass-calibration	JCGM 101:2008, 9.3, Tables 6 and 7	7	0.97	passed	1a32bc44777b
m3003-k4-weight-10kg	UKAS M3003 ed. 6 (2024), Appendix K.4	3	0.15	passed	ce1817c92061
m3003-k5-weighing-machine-200kg	UKAS M3003 ed. 6 (2024), Appendix K.5	3	0.50	passed	809b189d5416
nist-tn1900-e1-weighing	NIST Technical Note 1900, Example E1, Exhibit 1	6	0.27	passed	2d518dec2a7a
gum5-e14-radioactive-decay-bi207	JCGM GUM-5:2026, Example 14, 14.3 and Table 14.1	4	0.50	passed	6e332b0eb5d4
gum5-e15-wire-rope-breaking-force	JCGM GUM-5:2026, Example 15, 15.3, 15.4 and 15.5	5	0.67	passed	125859c53f87
iso5725-2-c2-softening-point-pitch-ref	ISO 5725-2:2019, Annex C.2.9 (Table C.13), Annex B.2	16	0.91	passed	9bf797a3de20
iso5725-2-c2-softening-point-pitch	ISO 5725-2:2019, Annex C.2 (Tables C.7–C.12, Formulae C.25–C.33)	31	1.0	passed	5ba888152956
koepke2017-half-life-cs137	Koepke, §7.2, Tables 3 and 4	13	0.98	passed	e50d5cde68e1
koepke2017-half-life-sr90	Koepke, §7.2, Tables 3 and 4	13	0.61	passed	20802cda308f
koepke2017-newtonian-constant-G	Koepke, §7.1, Tables 1 and 2	20	0.94	passed	63a4273da38f

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
merkatas2019-newtonian-constant-gas	Merkatas, Tables 1 and 4	14	0.69	passed	a1aae532511e
nicob-co60-activity-leave-one-out	NIST Census Builder user manual (Koepke, §4.4, Exhibits 19, 21 and 22; §A.5.1	68	0.82	passed	c4640bb263c5
nist-tn1900-e13-thermal-expansion-gauge-95	NIST Technical Note 1900, Example E13, Exhibit 20	6	0.89	passed	05abdaaa5acc
nist-tn1900-e13-thermal-expansion-gauge-99	NIST Technical Note 1900, Example E13, Exhibit 20	6	0.89	passed	23c7b16dfc57
nist-tn1900-e13-thermal-expansion-instrument-95	NIST Technical Note 1900, Example E13, Exhibit 20	3	0.69	passed	1d917adbd500
nist-tn1900-e13-thermal-expansion-instrument-99	NIST Technical Note 1900, Example E13, Exhibit 20	3	0.69	passed	361e0d5cd600
nist-tn1900-e3-falling-ball-viscometer	NIST Technical Note 1900, Example E3, Exhibit 3	7	0.55	passed	6920e13ab34d
num-11-stefan-boltzmann	T. Lafarge, §11, Table 5 and Figure 7	6	0.52	passed	fb04bea6cca7
possolo-meija-2022-hubble-lemaitre-constant	Possolo & Meija, Census Building, Hubble-Lemaître Constant, pp. 146–151	3	0.36	passed	6be95d5d53cd
emue-e2.2-vacuum-gauge-correction-applied	EMPIR 17NRM05 EMUE, Example E2.2, Tables E2.2.1–E2.2.3	4	0.90	passed	19e0b54745a0
emue-e2.2-vacuum-gauge-correction-ignored	EMPIR 17NRM05 EMUE, Example E2.2, Tables E2.2.1–E2.2.7	4	0.92	passed	1f42fa6f91e2
emue-e2.2-vacuum-gauge-correction-uncertainty	EMPIR 17NRM05 EMUE, Example E2.2, Tables E2.2.1–E2.2.5	4	0.92	passed	5a5fd2d10239
emue-e2.2-vacuum-gauge-sum-model-corrected	EMPIR 17NRM05 EMUE, Example E2.2, E2.2.7	4	0.068	passed	861d1957021f
emue-e2.2-vacuum-gauge-sum-model-ignored	EMPIR 17NRM05 EMUE, Example E2.2, E2.2.7	4	0.013	passed	08bfc3dacf84
m3003-k8-pressure-indicator-2mpa	UKAS M3003 ed. 6 (2024), Appendix K.8	3	0.45	passed	b24083f194e4
ea402-s11-block-calibrator-180d	EA-4/02 M:2022, S11	2	0.58	passed	623d8617c638
ea402-s5-thermocouple-emf	EA-4/02 M:2022, S5	3	0.67	passed	8f1a7a902650
ea402-s5-thermocouple-furnace	EA-4/02 M:2022, S5	3	0.37	passed	63cb7c26ccea
jcgm102-9.5.2-resistance-thermometer	JCGM 102:2011, 9.5.2 (Tables 12, 13)	2	0.59	passed	be62be046ce0
jcgm102-9.5.3-ten-temperatures	JCGM 102:2011, 9.5.3 (Tables 14–16)	31	0.96	passed	d48d618512c8
m3003-k11-gas-temperature	UKAS M3003 ed. 6 (2024), Appendix K.11 (K.11.17–K.11.18)	3	0.75	passed	fba975668f0d
m3003-k11-probe-calibration	UKAS M3003 ed. 6 (2024), Appendix K.11 (K.11.9)	2	0.51	passed	5d2049923f16
m3003-k7-thermocouple-emf	UKAS M3003 ed. 6 (2024), Appendix K.7	3	0.75	passed	d1eb52c4815e
m3003-k7-thermocouple-furnace	UKAS M3003 ed. 6 (2024), Appendix K.7 (K.7.19)	2	0.26	passed	0e0a20de7c1f
nist-tn1900-e2-surface-temperature	NIST Technical Note 1900, Example E2, Exhibit 2	6	0.98	passed	f51b0acd8aa8

The column max |Δ|/tol gives the largest ratio of the difference between computed and reference value to its tolerance in the file (≤ 1 passes). The tolerances reflect the rounding of the published values; for Monte Carlo results they add about five times the dispersion of the result over seeds, so that the comparison holds for other seeds as well (research log D15, D110).

Checked by: _____ Date: _____ Signature: _____

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